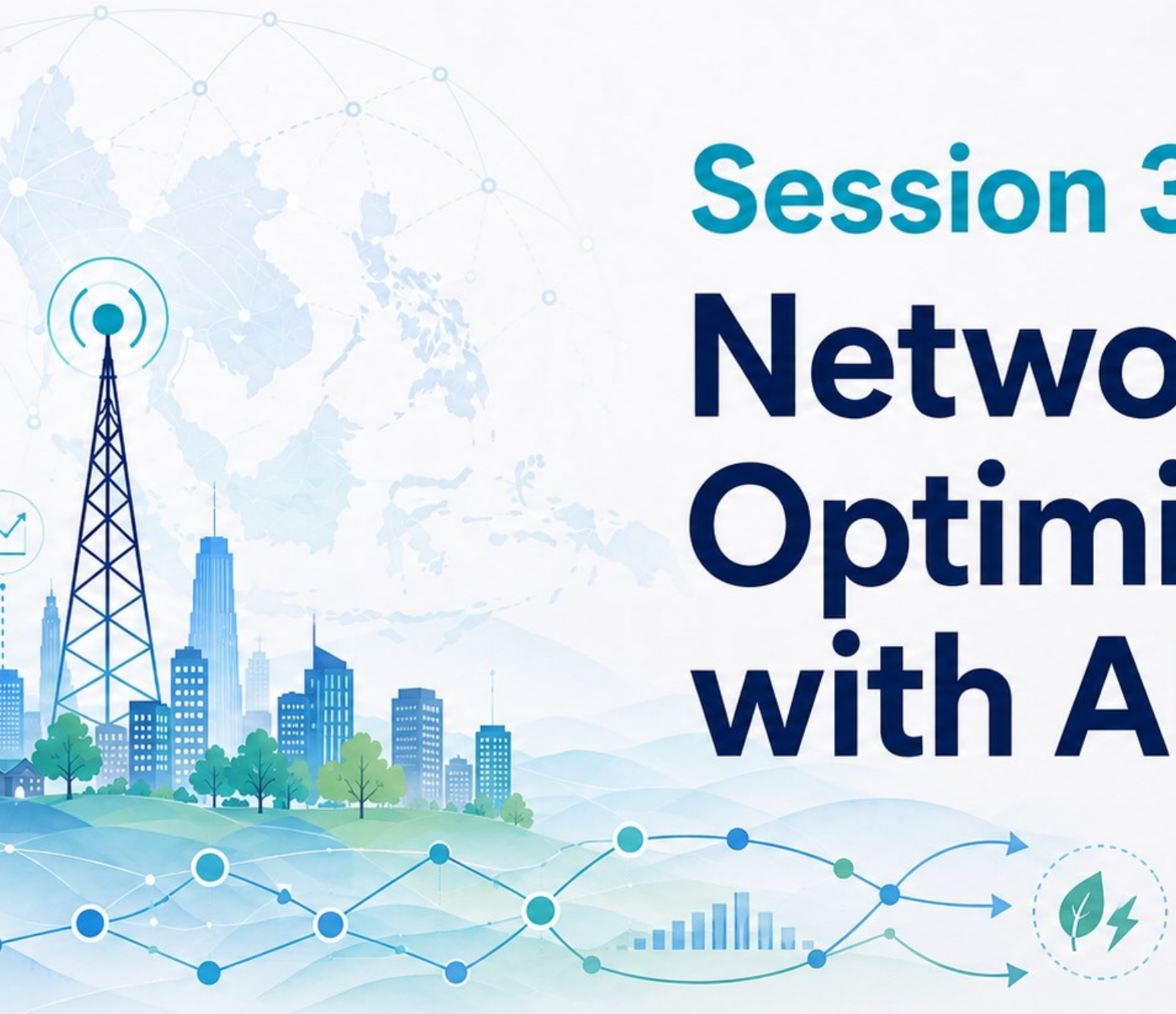
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GEOAI FUSION 2026

Module 7: Advanced Knowledge: Advanced Geospatial Techniques in GeoAI Applications (AD1002)

AGAIF 2026 Boot Camp 2
Level: Advanced Knowledge

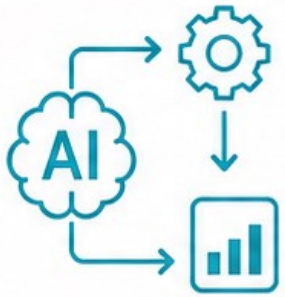
23 July 2026



Session 3:

Network Optimisation with AI

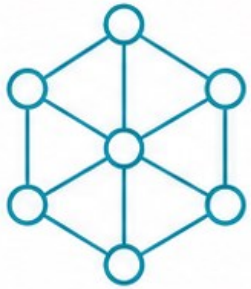
Learning Outcomes



1. Explain how AI supports network optimisation after a time-series forecast is produced.



2. Describe load balancing, dynamic spectrum allocation, energy efficiency and predictive maintenance.



3. Explain why a network can be represented as a graph and analysed with a Graph Neural Network (GNN).



4. Connect temporal prediction, spatial location and network relationships in a GeoAI workflow.

Session Agenda



1. Load balancing optimisation
2. Dynamic spectrum allocation
3. Energy efficiency analysis
4. Predictive maintenance
5. Graph Neural Networks for spatial and network-based GeoAI analysis



From Prediction to Network Optimisation



1. In Session 2, we used previous network records to predict the next condition.

2. For example, the LSTM predicted the next RSRP value for a specific time and location.

3. However, prediction alone does not improve the network.

4. The network must decide what action to take.

5. Session 3 explains how AI uses predicted and current network information to optimise network operations.

Session 2 predicts what may happen → **Session 3** decides what the network should do

Load Balancing Optimisation

What is load balancing?

Load balancing means sharing users and traffic across available cells so that no single cell becomes too busy.

Simple idea: move selected users from a busy cell to a cell with spare capacity.

Simple example



Cell A

Predicted load: 92%

RSRP: −92 dBm

Congested

Move selected users



Cell B

Predicted load: 55%

RSRP: −84 dBm

Spare capacity

−84 dBm is stronger than −92 dBm because it is less negative.

What does AI check before moving a user?



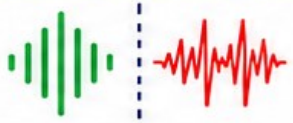
Predicted load

How busy will the cell be?



RSRP

How strong is the signal?



SINR

How clear is the signal?



User speed

Is the user moving fast or slowly?



Available PRBs

How many radio slots are free?

AI combines all five inputs to choose the best cell for the user.

What success looks like



Lower peak utilisation and fewer congested cells.



Higher or more stable user throughput.



Lower delay and fewer dropped sessions.



Fairer distribution of resources without repeated handovers.

Dynamic Spectrum Allocation

Give radio channels or bandwidth where they are needed most — while avoiding interference.

Scenario: Lunchtime in a business district

1



12:30 pm — Cell A becomes busy as many people use mobile data.

2



AI predicts the traffic peak and gives Cell A more bandwidth.

3



Cell B has lower demand, so it keeps less bandwidth. Different channels reduce interference.



Lunchtime

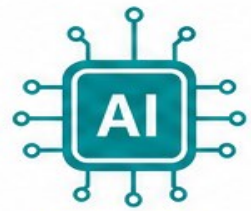


Busy Cell A

High demand

More bandwidth

Reassign bandwidth



AI Controller

Reassign bandwidth



Quiet Cell B

Low demand

Less bandwidth



Use different channels when signals interfere.

How the decision is made

1



Predict demand

Which cell will be busy?

2



Check channels

What is free and interference-safe?

3



Allocate spectrum

Give more capacity to high-demand cells.

4



Monitor & update

Adjust when network conditions change.



Simple idea: the right bandwidth, at the right cell, at the right time.

Energy Efficiency Analysis

Deliver useful communication while using as little energy as possible.

Dynamic spectrum allocation
Adjust channels and bandwidth

→

Same AI input: predicted network demand

→

Energy efficiency
Adjust power and sleep/wake states

Both optimise network resources while maintaining coverage, throughput and acceptable delay.

What does energy efficiency mean?

$$\text{Energy efficiency} = \frac{\text{Useful data delivered}}{\text{Energy consumed}}$$

10110
01001

- **bit/Joule** – communication efficiency
- **kWh** – total electricity use

More data with less energy = better efficiency.

Example: Cell C at night

1. 01:00 — Low demand predicted
2. AI checks nearby cells
3. 01:00–05:00 — Cell C uses low-power mode
4. Before demand rises — Cell C wakes up

Cell A · Cell B · Cell C

Forecast demand → Check coverage and quality → Reduce power safely → Wake before the next peak

Optimisation objective

Minimise total power consumption while maintaining service quality.

- Coverage maintained
- Minimum throughput met
- Delay limit respected

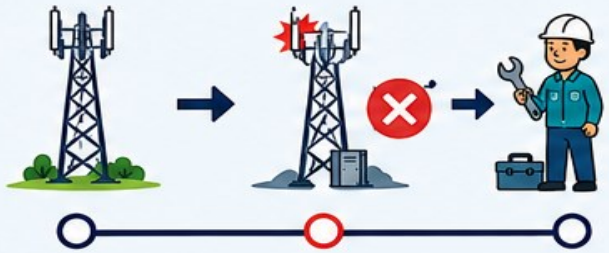
What Is Predictive Maintenance?

Definition: Predictive maintenance uses equipment data to detect possible problems before equipment fails.



Monitor equipment → Detect early warning → Plan maintenance → Avoid unexpected failure

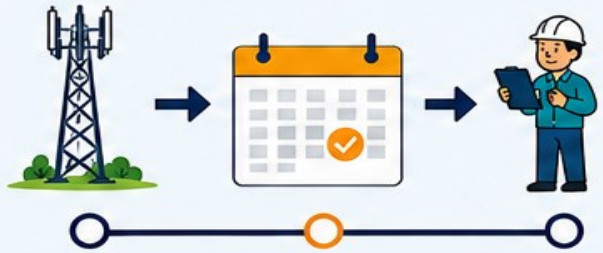
Reactive Maintenance



Repair **AFTER** equipment fails.

Unexpected outage may occur.

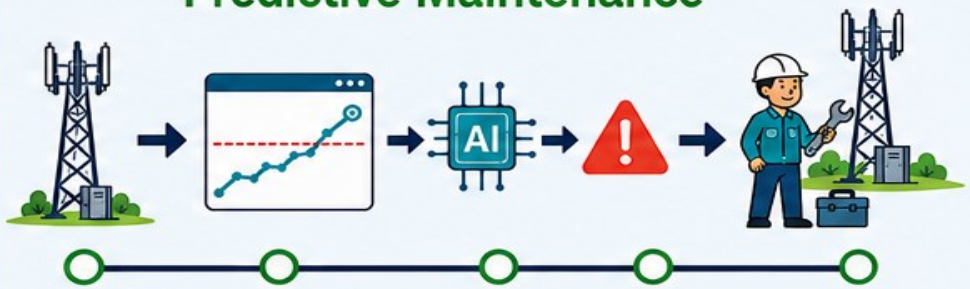
Preventive Maintenance



Maintain equipment on a **FIXED SCHEDULE.**

Maintenance may be done even when it is not yet needed.

Predictive Maintenance



Maintain equipment **WHEN DATA SHOWS** a possible problem.

Act early and reduce unnecessary maintenance.

Best use of condition data

Energy efficiency predicts demand; predictive maintenance predicts equipment health.

Simple idea: fix the equipment before it fails – based on evidence from data.

What Data Does AI Monitor?

Predictive maintenance acts before failure. To do this, AI needs evidence from equipment and network data.

Equipment condition

- Temperature
- Battery health
- Cooling fan
- Power use

Equipment history

- Past alarms
- Maintenance records

Cell C



Network performance

- Signal quality
- Packet loss
- Network errors

Environment

- Weather
- Outside temperature

Why does AI need several measurements?

Temperature rising	Power use rising	Repeated alarms
		

LSTM analyses temperature, power use and alarms over time to predict a possible equipment problem.



Example AI model

Possible equipment problem

Maintenance inspection needed

One unusual reading may be temporary. Several warning signs changing together over time provide stronger evidence of a problem.

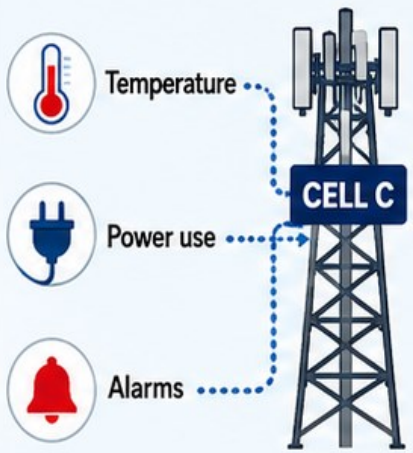
Detect problems earlier | Plan maintenance | Reduce unexpected outages

Data provides the evidence that predictive maintenance needs.

Predictive Maintenance Workflow

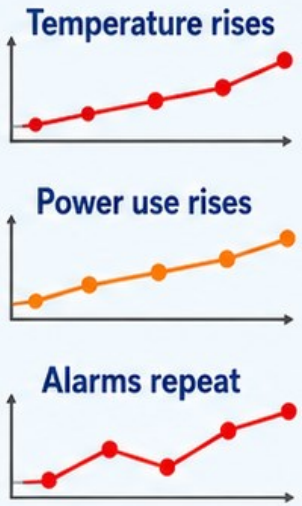
After AI monitors the data, it follows five simple steps.

1 Collect data



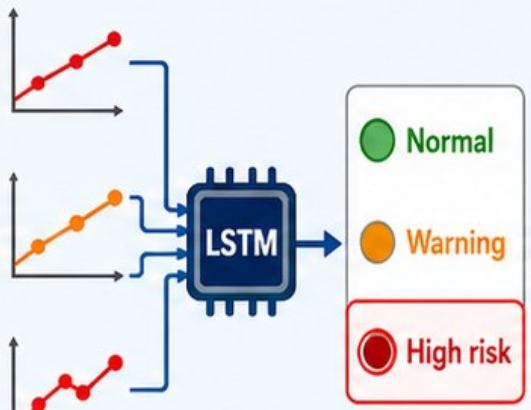
Cell C normally operates at **45°C**.

2 Notice a change



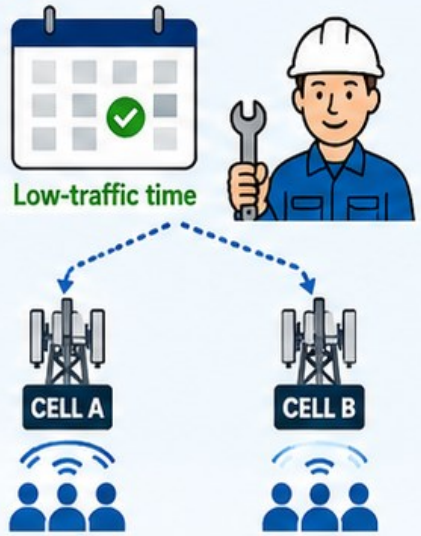
The pattern is different from normal.

3 Predict the risk



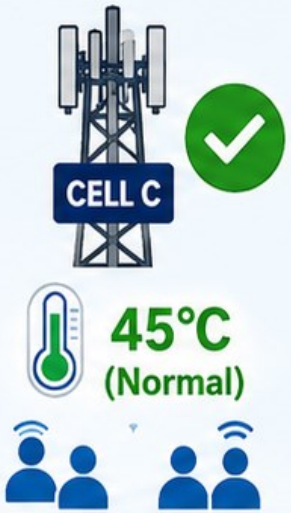
AI predicts a possible cooling problem.

4 Plan maintenance



Inspect Cell C before it fails.

5 Check the result



Temperature is normal and no outage occurs.

Without prediction:
Cell C may fail unexpectedly.

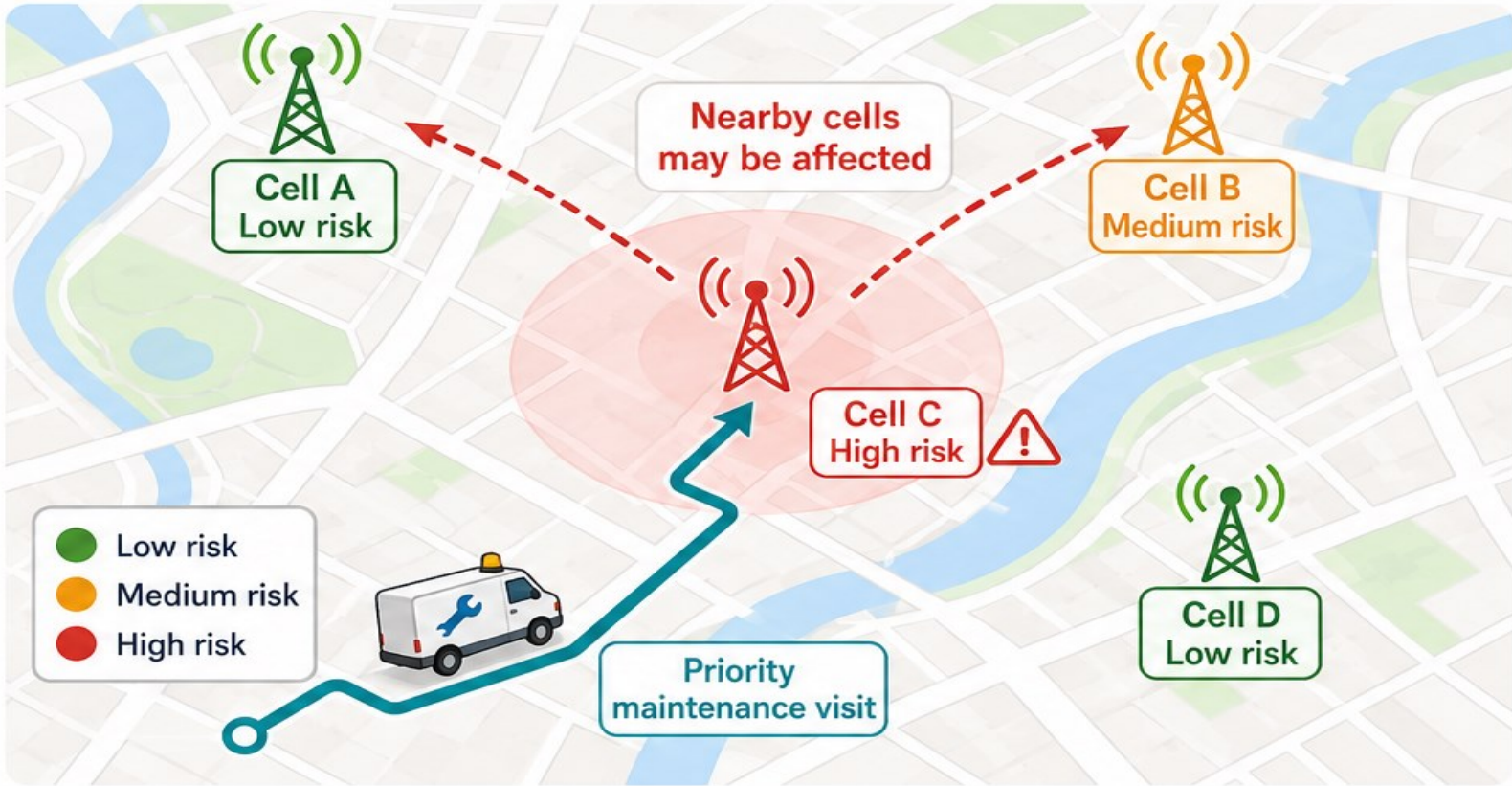
VS

With prediction:
Maintenance is completed before failure.

Collect data → detect change → predict risk → maintain early → avoid outage.

From Predictive Maintenance to GeoAI

Predictive maintenance becomes GeoAI when failure risk is linked to equipment location.



What does GeoAI add?

1. Map each cell tower
2. Show its failure-risk level
3. Identify nearby affected cells
4. Send maintenance teams to the highest-risk location

WHEN will demand change?
Time-series prediction

→ **WHERE** is the equipment?
GeoAI mapping

→ **WHICH** equipment may fail?
Predictive maintenance

Together, time, location and equipment condition support better network decisions.

GeoAI shows what may fail, where it is located and what nearby cells may be affected.

What's Next? From Predictive Maintenance to GNNs

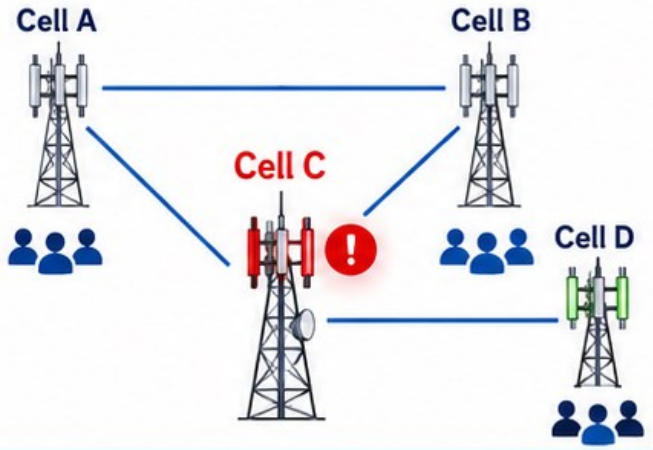
Predictive maintenance identifies equipment at risk. A GNN explains how that risk may affect the connected network.

1 PREDICT EQUIPMENT RISK



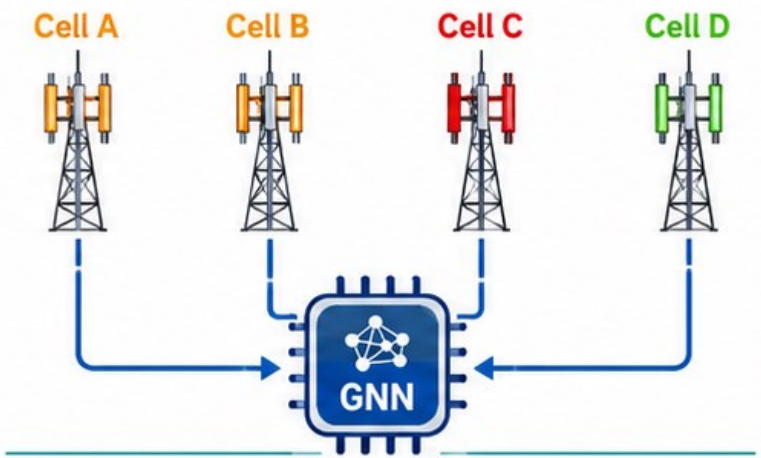
Cell C may fail.
Predictive maintenance detects the warning signs.

2 NETWORK EQUIPMENT IS CONNECTED



A cell does not operate alone.
A fault may affect neighbouring cells, links and users.

3 GNN ANALYSES THE WIDER IMPACT



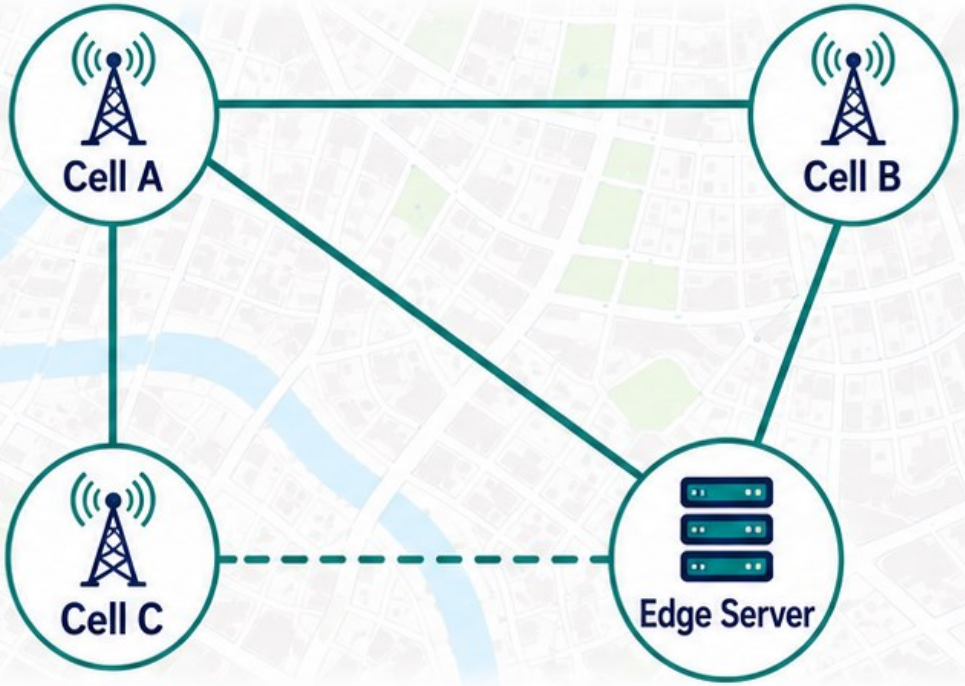
Which equipment is at risk?
Which connected cells may be affected?
Where should action be prioritised?

Predictive maintenance finds the possible fault → GNN analyses its impact across the network.

Next topic: Graph Neural Networks for spatial and network-based GeoAI analysis.

Graph Neural Networks for Spatial and Network-Based GeoAI

Why a graph? A network contains connected cells, users and links—not independent rows.



NODE = a network element at a location

EDGE = a connection between elements

1. NODE

- Cell tower, user or server
- GeoAI: an entity with a location

2. EDGE

- Link, handover, proximity or interference
- GeoAI: a spatial or network relationship

3. FEATURES

- RSRP, traffic load, energy, alarms, distance and link quality
- Data describing nodes and edges

4. TARGET

- Predict future load, allocation or failure risk
- The output for a node, edge or network

Simple idea: A graph represents WHERE network elements are and HOW they are connected.

How Does a GNN Work? — A Simple Example

A GNN checks one cell together with its connected neighbours before making a prediction.

1 CELL C HAS A PROBLEM

Cell C

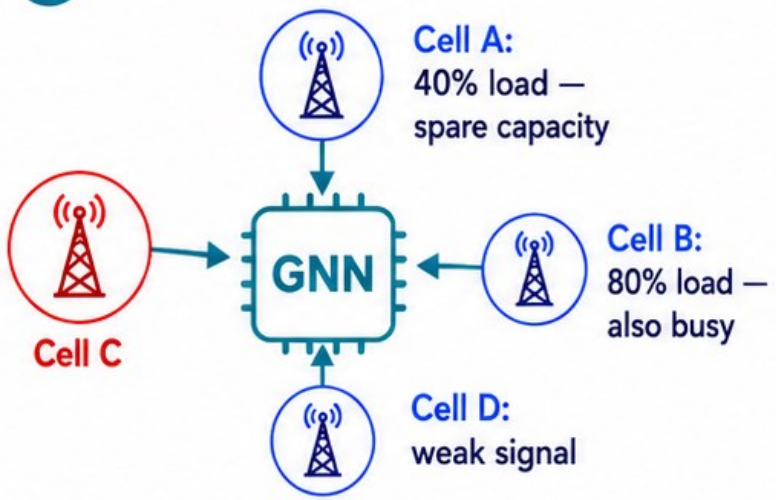
Load: 85%

RSRP: −90 dBm

Cell C may become congested.

Can another cell help?

2 GNN CHECKS CONNECTED CELLS



The GNN compares Cell C with cells that are connected to it.

This information sharing inside the model is called message passing.

3 GNN SUGGESTS AN ACTION



Best option: Cell A

Why? Lower load and suitable connection

Action: Move selected users to Cell A

The network operator or control system makes the final action.

Simple idea: GNN asks the neighbours before deciding.

Without a GNN: Cell C is checked alone.

With a GNN: Cell C and its connected neighbours are checked together.

LSTM and GNN Are Complementary

They answer different questions and can work together.

LSTM — TIME

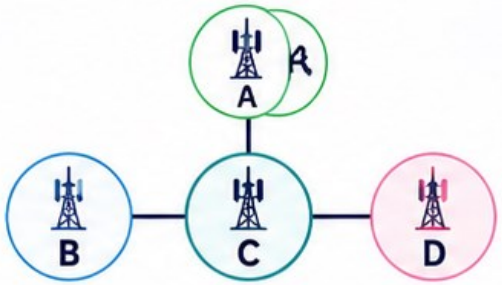


Main question:
How will this cell change over time?

Strength:
LEARNS TEMPORAL PATTERNS

Example:
Predict Cell C's future traffic load.

GNN — CONNECTIONS

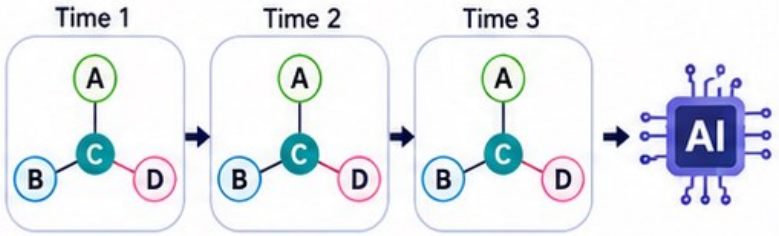


Main question:
How do connected cells affect one another?

Strength:
LEARNS NETWORK AND SPATIAL RELATIONSHIPS

Example: Check whether neighbouring cells can accept traffic.

TEMPORAL GNN — TIME + CONNECTIONS



Main question: How does the connected network change over time?

Strength: COMBINES TIME AND NETWORK TOPOLOGY

Example: Predict congestion across several connected cells.

PREDICT CONGESTION

Use past traffic and neighbouring-cell patterns.

OPTIMISE RESOURCES

Allocate power, bandwidth or channels across connected cells.

DETECT POSSIBLE FAULTS

Find abnormal cells or links that may affect the wider network.

LSTM answers WHEN; GNN explains WHAT IS CONNECTED; Temporal GNN combines both.

Summary of Session 3

Session 2 predicts what may happen; Session 3 decides what the network should do.

STAGE 1 — PREDICT WHAT HAPPENS NEXT



- **Cell A** predicted load: 92%
- **Cell B** predicted load: 55%
- **Cell C** predicted demand: low at night

TIME SERIES — **WHEN** will conditions change?

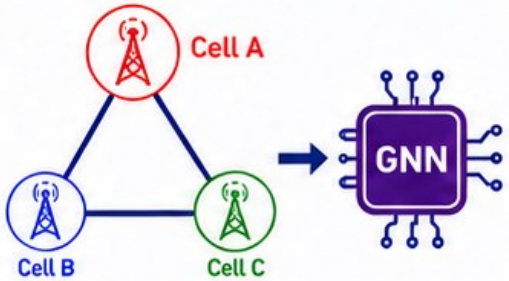
STAGE 2 — ADD LOCATION



GeoAI shows where each cell is located.

MAP — **WHERE** is the problem?

STAGE 3 — CHECK CONNECTIONS



GNN checks which cells are connected and how one cell may affect another.

GRAPH — **WHAT IS CONNECTED**?

WHAT IS THE NETWORK PROBLEM?

1 CONGESTION

Cell A is too busy

LOAD BALANCING
Move selected users to Cell B

2 INTERFERENCE OR LIMITED CAPACITY

Channel quality is poor

DYNAMIC SPECTRUM ALLOCATION
Change channel or assign more bandwidth

3 VERY LOW DEMAND

Cell C is quiet at night

ENERGY EFFICIENCY
Use a safe low-power state

4 EQUIPMENT WARNING

Temperature, power use and alarms are rising

PREDICTIVE MAINTENANCE
Inspect Cell C before failure

Monitor RSRP, throughput, delay, energy use and alarms after the action.

These actions do not always happen in sequence. AI selects the action that matches the predicted condition.

LSTM answers WHEN → GeoAI answers WHERE → GNN explains CONNECTIONS → optimisation selects the ACTION

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026